

Practical - 12

1. Producer Consumer

```
harshika@siri:~$ gedit producer_consumer.c
** (gedit:5235): WARNING **: 09:53:42.420: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:5235): WARNING **: 09:53:42.422: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc -Wall -pthread producer_consumer.c -o producer_consumer
harshika@siri:~$ ./producer_consumer
Consumer received: Data 1 from Producer
Consumer received: Data 2 from Producer
Consumer received: Data 3 from Producer
Consumer received: Data 4 from Producer
Consumer received: Data 5 from Producer

Communication time: 528.00 microseconds
harshika@siri:~$ ./producer_consumer 10
Consumer received: Data 1 from Producer
Consumer received: Data 2 from Producer
Consumer received: Data 3 from Producer
Consumer received: Data 4 from Producer
Consumer received: Data 5 from Producer

Communication time: 712.00 microseconds
harshika@siri:~$ gedit producer_consumer.c
** (gedit:5440): WARNING **: 09:54:43.217: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:5440): WARNING **: 09:54:43.218: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/wait.h>
5 #include <sys/time.h>
6 #include <string.h>
7 int main() {
8     int pipefd[2];
9     pid_t pid;
10    char buffer[100];
11    struct timeval start, end;
12    double time_taken;
13    if (pipe(pipefd) == -1) {
14        perror("pipe");
15        exit(1);
16    }
17    pid = fork();
18    if (pid < 0) {
19        perror("fork");
20        exit(1);
21    }
22    if (pid > 0) {
23        close(pipefd[0]);
24        gettimeofday(&start, NULL);
25        for (int i = 1; i <= 5; i++) {
26            sprintf(buffer, "Data %d from Producer\n", i);
27            write(pipefd[1], buffer, strlen(buffer));
28        }
29        close(pipefd[1]);
30        wait(NULL);
31        gettimeofday(&end, NULL);
32        time_taken = (end.tv_sec - start.tv_sec) * 1000000.0;
33        time_taken += end.tv_usec - start.tv_usec;
34        printf("\nCommunication time: %.2f microseconds\n", time_taken);
35    }
36    else {
37        close(pipefd[1]);
38        char ch;
39        int index = 0;
40        while (read(pipefd[0], &ch, 1) > 0) {
41            if (ch == '\n') {
42                buffer[index] = '\0';
43                printf("Consumer received: %s\n", buffer);
44                index = 0;
45            }
46            else {
47                buffer[index++] = ch;
48            }
49        }
50        close(pipefd[0]);
51        exit(0);
52    }
53    return 0;
54 }
```

2. Deadlock

```
harshika@siri:~$ gedit deadlock_demo.c
** (gedit:5652): WARNING **: 09:56:26.865: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:5652): WARNING **: 09:56:26.865: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
harshika@siri:~$ gcc -Wall -pthread deadlock_demo.c -o deadlock_demo
harshika@siri:~$ ./deadlock_demo
=====
DEADLOCK DEMONSTRATION
=====

Thread 1: Trying to lock Resource 1...
Thread 1: Locked Resource 1.
Thread 2: Trying to lock Resource 2...
Thread 2: Locked Resource 2.
Thread 2: Trying to lock Resource 1...
Thread 1: Trying to lock Resource 2...
^C
harshika@siri:~$ gedit deadlock_demo.c
** (gedit:6050): WARNING **: 09:57:42.810: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:6050): WARNING **: 09:57:42.811: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found

1 #include <stdio.h>
2 #include <pthread.h>
3 #include <unistd.h>
4 pthread_mutex_t resource1;
5 pthread_mutex_t resource2;
6
7 void *thread1_function(void *arg)
8 {
9     printf("Thread 1: Trying to lock Resource 1...\n");
10    pthread_mutex_lock(&resource1);
11    printf("Thread 1: Locked Resource 1.\n");
12    sleep(1);
13    printf("Thread 1: Trying to lock Resource 2...\n");
14    pthread_mutex_lock(&resource2);
15    printf("Thread 1: Locked Resource 2.\n");
16    printf("Thread 1: Using both resources.\n");
17    pthread_mutex_unlock(&resource2);
18    pthread_mutex_unlock(&resource1);
19    return NULL;
20 }
21
22 void *thread2_function(void *arg)
23 {
24     printf("Thread 2: Trying to lock Resource 2...\n");
25    pthread_mutex_lock(&resource2);
26    printf("Thread 2: Locked Resource 2.\n");
27    sleep(1);
28    printf("Thread 2: Trying to lock Resource 1...\n");
29    pthread_mutex_lock(&resource1);
30    printf("Thread 2: Locked Resource 1.\n");
31    printf("Thread 2: Using both resources.\n");
32    pthread_mutex_unlock(&resource1);
33    pthread_mutex_unlock(&resource2);
34    return NULL;
35 }
36
37 int main()
38 {
39 }
```